Lecture 1

- State estimation intro
- Syllabus overview
- Probability and random variables review

Q: What is state estimation?

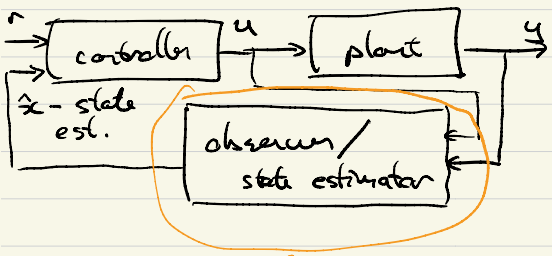
Consider control system:

continuous-time:

$\dot{x} = f(x,u)$ ← dynamics function — x - state $\in \mathbb{R}^n$

$y = g(x,u)$ ← measurement function — u - output $\in \mathbb{R}^m$, y - output $\in \mathbb{R}^p$

recall feedback control:



↑ this class is all about this module

For this class instead we consider discrete-time models:

$x_{t+1} = f(x_t, u_t)$ (noise-free model)

$y_t = g(x_t, u_t)$

- tends to give simpler theory
- more natural for algorithm implementation

E.g. Linear Time-Invariant state-space model (LTI)

$x_{t+1} = Ax_t + Bu_t$

$y_t = Cx_t + Du_t$

- for this kind of model, the Kalman Filter is "optimal" in a sense we will see later.

Goal:

Given: system model $f(\cdot,\cdot)$, $g(\cdot,\cdot)$ and noise statistics, e.g. Q, R.

Given: input sequence $u_{0:t-1} := (u_0, u_1, \dots, u_{t-1})$

output sequence $y_{1:t} := (y_1, y_2, \dots, y_t)$

Objective: estimate state at time t, $\hat{x}_t$ "close to" x_t (x_t unknown).

Q: Why is this not just an observer?

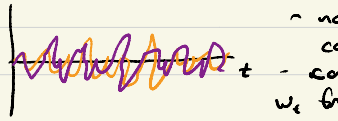
A: Because noise (randomness) makes it a challenge!

Really our model:

$x_{t+1} = f(x_t, u_t) + w_t$ ← process noise

$y_t = g(x_t, u_t) + v_t$ ← sensor noise

Typically (w_t, v_t) - white noise processes

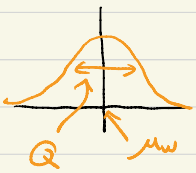


- no time correlation
- cannot predict w_t from w_τ ($\tau < t$)

- white noise processes are specified by their statistics

E.g. Gaussian White Noise

$w_t \sim \mathcal{N}(\mu_w, Q)$ — Gaussian dist.; Q: covariance matrix; μ_w: mean, usually 0



- "big" Q means large magnitude w_t, v_t means lots of randomness in $f(\cdot,\cdot)$, $g(\cdot,\cdot)$.

similarly $v_t \sim \mathcal{N}(\mu_v, R)$ — covariance; μ_v usually 0

Probabilistic Perspective

consider an unknown random process:

$$X_{0:t} = (X_0, X_1, \ldots, X_t) \sim p(x_{0:t})$$

$$Y_{0:t} = (Y_0, Y_1, \ldots, Y_t) \sim p(y_{0:t})$$

↑ these we can measure, we have realization: $y_{0:t}$

assume $Y_{0:t}$, $X_{0:t}$ are correlated, with joint dist.

$$p(x_{0:t}, y_{0:t})$$

we want to use $Y_{0:t}$ to predict $X_{0:t}$ by computing posterior distribution:

posterior:

$$p(x_{0:t} \mid y_{0:t})$$

$$\hookrightarrow p(x_t \mid y_{0:t})$$

↑ dist. over state at time t

we might take $\hat{x}_t$ to be max or mean of this dist.

Review of Probability and Random Variables

X - random unknown quantity a.k.a. a Random Variable (RV)

x - a realization, specific value of that RV

we say $X \sim p(x)$ — probability distribution function

x takes on values in some domain $\mathcal{D}$

$x \in \mathcal{D}$

E.g. $x \in \mathbb{R}$ real valued scalar

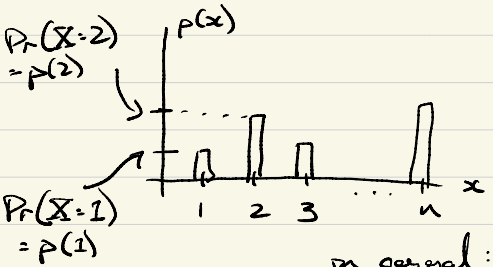
$x \in \mathbb{R}^n$ real valued n-dim vector

$x \in \{0,1\}$ binary R.V.

$x \in \{1, 2, \ldots, n\}$ discrete R.V.

the distribution $p(x)$ describes the probability of X taking on a value $x \in \mathcal{S} \subset \mathcal{D}$ (any subset of $\mathcal{D}$).

E.g. Discrete R.V.



in general:

$$\Pr(X \in \mathcal{S}) = \sum_{x \in \mathcal{S}} p(x)$$

E.g. $\Pr(X \in \{1,2\}) = p(1) + p(2)$

also:

$p(x)$ is always such that

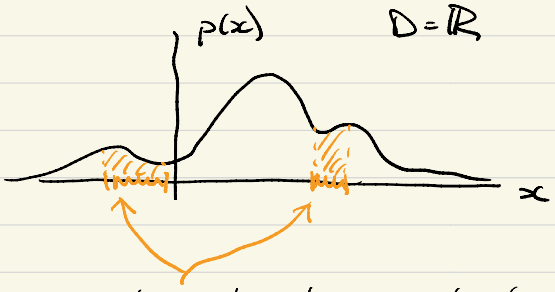
$$\sum_{x \in \mathcal{D}} p(x) = 1$$
$$0 \le p(x) \le 1$$

for discrete R.V. we call $p(x)$ a probability mass function (PMF)

E.g. Continuous Real-Valued R.V.s

let $X \sim p(x)$

$D = \mathbb{R}$



let this be a subset $S \subset \mathbb{R}$

$$P_r(X \in S) = \int_{x \in S} p(x)\, dx$$

- for cont. R.V. we call $p(x)$ a probability density function (PDF).
- can also think of $p(x)$ as a differential probability mass.